

Information-Theoretic Multi-Agent Path Planning for Risk-Aware Navigation Under Environmental Uncertainty

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Autonomous navigation in severe weather requires balancing information gathering and mission safety under extreme uncertainty. A novel multi-agent collaborative framework addresses this challenge through information-theoretic path planning that coordinates High-Value Aerial Assets (HVAA) with expendable Sensing Aerial Assets (SAA) in hurricane-like conditions. The framework employs modified Rapidly-exploring Random Tree star (RRT*) algorithms: Expected Travel Time RRT* for conservative HVAA planning and Information-Theoretic RRT* for SAA exploration that maximizes uncertainty reduction. To guide SAA behavior, an entropy-weighted utility function jointly optimizes travel cost and expected information gain, enabling targeted sampling of high-uncertainty regions characterized by bimodal wind distributions. A nonlinear trajectory-tracking controller with look-ahead guidance ensures feasible execution of planned path under strong wind disturbances. Hurricane Harvey 2017 wind data validates the approach, demonstrating 28% path length reduction and 24.1% travel time improvement compared to baseline non-collaborative planning. Under moderate wind conditions (6-8 m/s), the system achieves 2.0-meter average tracking accuracy with 32-meter maximum deviation during aggressive maneuvers. Performance analysis reveals clear operational boundaries, with acceptable operation below 14 m/s wind speeds and graceful degradation beyond these limits. The framework enables safer autonomous navigation in uncertain environments while quantifying operational constraints for deployment guidance.

I. Nomenclature

A	=	amplitude of oscillation
a	=	cylinder diameter, m
$\mathbf{a}_{NM}$	=	normal acceleration command, m/s^2
C_p	=	pressure coefficient
C_x	=	force coefficient in the x direction
C_y	=	force coefficient in the y direction
d	=	shift distance in look-ahead guidance, m
$d_{i,j}$	=	distance between cells i and j , m
d_{shift}	=	boundary layer shift distance, m
dt	=	time step, s

$\mathbf{e}(t)$	= tracking error vector, m
$ETT_{i,p}$	= expected travel time for agent i , path p , s
F_x	= x component of resultant pressure force, N
F_y	= y component of resultant pressure force, N
G_{collab}	= collaborative gain factor
h	= height, m
$H(x)$	= Shannon entropy for variable x
i	= agent index or cell index
J	= total integrated cost
j	= cell or waypoint index
k	= guidance gain
K	= trailing-edge nondimensional angular deflection rate
$\hat{\mathbf{L}}(t)$	= look-ahead vector
N	= number of RRT* iterations or total number of cells
$P_{j,y}$	= probability of cell j being type y
q	= configuration in RRT* tree
R	= influence radius for spatial correlation, m
$\mathbf{r}_M(t)$	= vehicle position vector, m
$\mathbf{r}_P(t)$	= closest point on desired path, m
t	= time, s
$T_{j,y}$	= travel time through cell j under condition y , s
$\hat{\mathbf{T}}_P$	= unit tangent vector of desired path
$U_{p,t}^{\text{SAA}}$	= SAA utility function for path p at time t
$U_{p,t}^{\text{HVAA}}$	= HVAA utility function for path p at time t
v_M^*	= maximum airspeed, m/s (14 m/s)
$\mathbf{v}_A$	= airspeed velocity vector, m/s
$\mathbf{v}_{\text{ground}}$	= ground speed velocity vector, m/s
$\mathbf{v}_M$	= vehicle velocity vector, m/s
$\mathbf{w}$	= wind velocity vector, m/s
w^*	= wind speed magnitude, m/s
w_a	= control effort weight (0.1)
w_e	= tracking error weight (1.0)
$W_{i,1}$	= information gain weight for agent i
$W_{i,2}$	= path cost normalization weight for agent i
Y	= number of mixture components or cell types
β	= wind feasibility parameter, rad
δ_{BL}	= boundary layer width, m
η_{info}	= information utilization efficiency
θ	= angular difference between path and wind, deg
Θ_{max}	= maximum allowable angular difference, deg (50°)
θ_L	= look-ahead angle, rad
θ_{s2}	= rotation angle for curvature compensation, rad
κ_P	= path curvature, m^{-1}
λ	= wind angle relative to desired direction, rad
λ_j	= dynamic maneuvering cue (DMC) risk in cell j
$\lambda_{\text{max,HVAA}}$	= maximum allowable HVAA risk (0.75)
$\lambda_{\text{max,SAA}}$	= maximum allowable SAA risk (0.85)
$\mu_{j,y}$	= mean of wind distribution in cell j , component y , m/s
ρ	= density, kg/m^3
$\sigma_{j,y}$	= standard deviation of wind distribution in cell j , component y , m/s
$\sigma_{\text{max,HVAA}}$	= maximum allowable HVAA uncertainty (90)
$\sigma_{\text{max,SAA}}$	= maximum allowable SAA uncertainty (120)
$\sigma_{\text{posterior}}$	= posterior standard deviation after observation
σ_{prior}	= prior standard deviation before observation

$\omega(i, j)$	=	spatial correlation weighting function
DESPOT	=	Determinized Sparse Partially Observable Tree
DEZ	=	Dynamic Engagement Zone
DMC	=	Dynamic Maneuvering Cue
EnKF	=	Ensemble Kalman Filter
ETT	=	Expected Travel Time
GDA	=	Global Data Assimilation
HVAA	=	High Value Aerial Asset
IT	=	Information Theoretic
IT-RRT*	=	Information-Theoretic Rapidly-exploring Random Tree star
KL	=	Kullback-Leibler
MDP	=	Markov Decision Process
ML	=	Machine Learning
POMDP	=	Partially Observable Markov Decision Process
RMS	=	Root Mean Square
RRT*	=	Rapidly-exploring Random Tree star
RRT*-ETT	=	RRT* with Expected Travel Time
RRT*-IT	=	RRT* with Information Theoretic utility
SAA	=	Sensing Aerial Asset
SARSOP	=	Successive Approximations of the Reachable Space under Optimal Policies
SD	=	Standard Deviation
TMML	=	Temporal Multimodal Multivariate Learning
UAV	=	Unmanned Aerial Vehicle
avg	=	average value
collab	=	collaborative
emergency	=	emergency mode
i	=	agent index (HVAA or SAA)
info	=	information-related
j	=	cell or waypoint index
max	=	maximum value
min	=	minimum value
M	=	vehicle (manned or autonomous)
NM	=	normal (perpendicular to velocity)
P	=	path
posterior	=	after observation
prior	=	before observation
rand	=	random
y	=	distribution type or mixture component

II. Introduction

Autonomous aerial systems operating in hazardous environments must navigate complex trade-offs between mission efficiency and safety under extreme uncertainty. Hurricane reconnaissance, disaster response, and military operations in contested environments exemplify scenarios where environmental conditions create significant navigational challenges while simultaneously demanding optimal performance[1]. Traditional path planning approaches rely on conservative worst-case assumptions that often result in suboptimal routes, while aggressive strategies risk mission failure or asset loss when uncertainties materialize unfavorably.

The fundamental challenge lies in managing incomplete environmental information effectively. Weather prediction models provide forecasts with inherent uncertainty, particularly in rapidly changing conditions such as tropical cyclones where wind speeds and directions exhibit bimodal distributions due to complex atmospheric dynamics . Conventional single-agent path planning algorithms address uncertainty through conservative margin-based approaches or probabilistic methods that assume fixed uncertainty distributions throughout the mission duration .

Recent advances in multi-agent systems have demonstrated the potential for collaborative sensing to reduce environmental uncertainty [2, 3]. Cooperative approaches leverage expendable or lower-value assets to gather

information that benefits high-value platforms, enabling more informed decision-making. However, existing frameworks often treat sensing and navigation as separate optimization problems, failing to capture the coupled nature of information gathering and path planning under dynamic conditions. Information-theoretic approaches to robotics have shown promise in addressing exploration-exploitation trade-offs by quantifying the value of information gained through sensing actions. These methods typically employ entropy reduction or mutual information metrics to guide sensor placement and sampling strategies. However, their application to multi-agent path planning in severe weather environments remains limited, particularly when considering the temporal constraints and risk thresholds inherent in operational scenarios.

Current multi-agent path planning research has primarily focused on static or slowly-varying environments with well-characterized obstacles. Extensions to uncertain environments often rely on Partially Observable Markov Decision Processes (POMDPs) or belief space planning, which become computationally intractable for large-scale problems with continuous state spaces. Alternative approaches using sampling-based methods such as Rapidly-exploring Random Trees (RRT) have demonstrated scalability advantages but lack systematic frameworks for incorporating information-theoretic objectives in multi-agent scenarios.

The integration of vehicle dynamics constraints with uncertainty-aware path planning presents additional complexities. Strong wind disturbances significantly affect trajectory tracking performance, requiring controllers that can maintain path following accuracy while respecting actuator limitations. The coupling between path feasibility and environmental uncertainty creates a feedback loop where information gathering capabilities directly influence navigation performance.

A collaborative framework addresses these limitations through information-theoretic multi-agent path planning that explicitly coordinates sensing and navigation objectives. The approach employs differentiated agent roles: risk-averse High-Value Aerial Assets (HVAA) that prioritize mission completion within safety constraints, and expendable Sensing Aerial Assets (SAA) that maximize information gain to benefit team performance. Modified RRT* algorithms incorporate uncertainty quantification and information-theoretic utility functions, enabling systematic exploration of high-uncertainty regions while maintaining computational tractability.

The framework contributions include: Information-theoretic RRT* algorithms that balance path cost with uncertainty reduction objectives, collaborative utility functions that coordinate multi-agent sensing and navigation, nonlinear trajectory tracking integration that ensures path feasibility under strong wind disturbances, and comprehensive performance analysis quantifying operational boundaries and system limitations.

Section III formulates the multi-agent navigation problem under environmental uncertainty. Section IV details the information-theoretic RRT* methodology and collaborative planning algorithms. Section V presents simulation results using Hurricane Harvey wind data, demonstrating performance improvements and operational constraints. Section V concludes with key findings and future research directions.

III. Problem Formulation

This section establishes the mathematical framework for multi-agent collaborative navigation under environmental uncertainty. We first describe the environmental representation including bimodal wind distributions and dynamic threat zones[4]. The multi-agent system architecture defines roles and capabilities of the HVAA and SAA. Finally, we formulate objective functions that balance travel efficiency, risk avoidance, and information acquisition.

A. Environmental Model

The operational environment is discretized into a uniform grid of $N \times N$ cells, where each cell j is characterized by wind speed distribution parameters and threat level indicators. The wind velocity in cell j consists of a known directional component $\hat{w}_j$ and an uncertain magnitude w_j drawn from a probability distribution. For hurricane affected regions, atmospheric mixing phenomena create bimodal distributions representing distinct weather regimes within the same spatial location.

The wind speed distribution for cell j at time t is modeled as a Gaussian mixture:

$$p(w_j, t) = \sum_{y=1}^Y P_{j,y} \mathcal{N}(\mu_{j,y}, \sigma_{j,y}^2)$$

where Y represents the number of mixture components, $P_{j,y}$ denotes the probability of component y , and $\mathcal{N}(\mu, \sigma^2)$ indicates a Gaussian distribution with mean μ and variance σ^2 . The standard deviation $\sigma_{j,t}$ quantifies epistemic uncertainty arising from incomplete environmental knowledge. Regions with $\sigma_{j,t} < 40$ exhibit low uncertainty

with stable unimodal conditions, while $\sigma_{j,t} > 90$ indicates extreme uncertainty characteristic of severe atmospheric disturbances.

Dynamic threat zones represent time-varying engagement regions where aerial assets face detection and interdiction risks. The threat level in cell j is parameterized by the Dynamic Maneuvering Cue (DMC) risk metric λ_j , ranging from 0 (no threat) to 1 (certain engagement). The spatial distribution of threats evolves according to adversarial decision processes not explicitly modeled in this formulation.

Figure 1 illustrates a representative environment derived from Hurricane Harvey 2017 meteorological data. The uncertainty field exhibits characteristic cyclonic structure with σ values ranging from 20 near the periphery to 125 in the eyewall region centered at coordinates (82, 80). Overlaid wind vectors demonstrate the rotational flow pattern with velocities proportional to radial distance from the storm center until the eyewall, followed by exponential decay in the outer bands.

B. Multi-Agent System Architecture

The system comprises two heterogeneous aerial vehicles with distinct operational roles and capabilities. The High-Value Aerial Asset executes the primary mission objective of reaching a goal location while minimizing travel time subject to strict safety constraints. The Sensing Aerial Asset performs reconnaissance to reduce environmental uncertainty, enabling the HVAA to identify more efficient routing options.

1. High-Value Aerial Asset

The HVAA operates with conservative risk tolerance, avoiding regions where environmental uncertainty or threat levels exceed predefined thresholds. The vehicle maintains constant airspeed $v_M = 14$ m/s and can sustain operations across altitude ranges from 70,000 to 87,400 meters. Onboard sensing capabilities provide only local environmental observations insufficient for global path optimization. The HVAA relies on prior information from satellite imagery and regional weather forecasts, supplemented by reconnaissance data transmitted from the SAA.

Risk constraints for the HVAA are formulated as maximum allowable thresholds on both uncertainty and threat exposure:

$$\sigma_j \leq \sigma_{\max, \text{HVAA}}$$

$$\lambda_j \leq \lambda_{\max, \text{HVAA}}$$

where $\sigma_{\max, \text{HVAA}} = 90$ and $\lambda_{\max, \text{HVAA}} = 0.75$ represent operational limits beyond which mission safety cannot be assured. Paths violating these constraints are rejected during the planning process regardless of potential efficiency gains.

2. Sensing Aerial Asset

The SAA is characterized as an expendable platform with enhanced sensing capabilities and relaxed risk tolerance relative to the HVAA. The vehicle maintains identical airspeed $v_M = 14$ m/s but operates under less restrictive constraints ($\sigma_{\max, \text{SAA}} = 120$, $\lambda_{\max, \text{SAA}} = 0.85$) to enable exploration of moderately uncertain or threatened regions. Onboard sensors provide perfect environmental observations within visited cells, collapsing probability distributions to deterministic values. This assumption represents an idealized sensing model; practical implementations would incorporate measurement noise and partial observability.

Communication between agents occurs over a radio link with 200-mile maximum range and negligible latency. The SAA transmits environmental observations to the HVAA in real-time, enabling dynamic replanning as new information becomes available. Loss of communication triggers a graceful degradation protocol where the HVAA reverts to conservative planning based solely on prior information.

C. Objective Functions

1. Expected Travel Time

For a given path p traversing cell sequence $\{j_1, j_2, \dots, j_J\}$, the expected travel time accounts for uncertain wind effects on ground speed. The travel time through cell j depends on both the path-aligned wind component and the

probability distribution of wind magnitudes:

$$ETT_{i,p} = \sum_{j=1}^J \sum_{y=1}^Y P_{j,y} T_{j,y}$$

$P_{j,y}$ is the probability of cell j being of type y , T_y is the travel time through a cell of type y , and Y is the total number of cell types.

Three planning strategies are considered based on different interpretations of the bimodal distribution. Expected travel time planning uses the probability-weighted average from Eq. (3). Worst-case planning assumes maximum travel time: $ETT_{\text{worst}} = \sum_{j=1}^J \max_y (T_{j,y})$, providing conservative guarantees at the cost of potential over-estimation. Best-case planning assumes minimum times but offers no safety assurances.

2. Information Gain Quantification

Information gain is quantified through reduction in standard deviation when the SAA observes cell j . Under the complete uncertainty removal assumption, observation transforms the distribution from prior $\mathcal{N}(\mu_j, \sigma_j^2)$ to posterior $\mathcal{N}(\mu_j, 0)$, eliminating epistemic uncertainty. The information gained is proportional to the prior standard deviation:

$$IG_j = \sigma_{\text{prior},j} - \sigma_{\text{posterior},j} = \sigma_{\text{prior},j}$$

This formulation prioritizes exploration of highly uncertain regions where observations provide maximum value for reducing epistemic risk.

For bimodal distributions, observation additionally resolves which mode applies, collapsing the mixture to a unimodal distribution. If mode probabilities fall within $[0.4, 0.6]$, the realized mode is selected randomly. If probabilities are more extreme (outside $[0.3, 0.7]$), the most likely mode is selected deterministically.

3. Multi-Agent Utility Functions

The SAA utility function balances path cost against information gain through weighted summation:

$$U_{p,t}^{\text{SAA}} = W_{\text{SAA},2} \cdot \text{cost}_{\text{SAA},t} - \sum_{j=0}^{\text{end}} \left[W_{j,1} (\sigma_{\text{prior},j} - \sigma_{\text{posterior},j}) \right]$$

where $W_{\text{SAA},2}$ normalizes path cost and $W_{j,1}$ weights information gain based on entropy. The entropy-based weight prioritizes cells with high distributional uncertainty:

$$W_{j,1} = \log_2 \left[\frac{1}{\max(P_{j,t}(T_{\min,j,t}), 1 - P_{j,t}(T_{\min,j,t}))} \right]$$

This weighting ensures that cells with broad bimodal distributions (high entropy) receive greater exploration priority than narrow unimodal distributions (low entropy).

The HVAA utility function omits information gain terms, focusing solely on travel time minimization subject to safety constraints:

$$U_{p,t}^{\text{HVAA}} = W_{\text{HVAA},2} \cdot \text{cost}_{\text{HVAA},t}$$

The weight $W_{\text{HVAA},2}$ remains fixed throughout Thread 1; dynamic adjustment of this parameter to optimize collaborative timing is reserved for Thread 2 investigation.

Collaborative performance is measured through the information utilization efficiency

$$\eta_{\text{info}} = \frac{\text{information used}}{\text{information gathered}}$$

quantifying how effectively SAA reconnaissance data enables HVAA path improvements. A collaborative gain factor

$$G_{\text{collab}} = \frac{T_{\text{individual}}}{T_{\text{collaborative}}}$$

exceeding unity indicates net benefit from multi-agent coordination.

D. Vehicle Dynamics Constraints

Path feasibility depends critically on wind-relative kinematics. The angular difference θ between desired path segment direction $\hat{\mathbf{b}}_i$ and wind direction $\hat{\mathbf{w}}_i$ must satisfy energy-efficient navigation constraints:

$$\theta = \arccos\left(\frac{\hat{\mathbf{w}}_i \cdot \hat{\mathbf{b}}_i}{\|\hat{\mathbf{w}}_i\| \|\hat{\mathbf{b}}_i\|}\right) \leq \Theta_{\max}$$

where $\Theta_{\max} = 50^\circ$ represents the maximum acceptable angular deviation. This constraint ensures the vehicle can maintain forward progress without excessive control effort. Path segments violating Equation are rejected during tree construction, guaranteeing dynamic feasibility of all candidate paths.

IV. Methodology

This section presents the information-theoretic multi-agent path planning framework for collaborative navigation under environmental uncertainty. We first describe modifications to the standard RRT* algorithm that enable explicit information gain optimization. Conservative planning strategies for the HVAA minimize expected travel time while enforcing strict safety constraints. Exploratory planning for the SAA balances information acquisition with path cost through entropy-weighted utility functions. Finally, we integrate vehicle dynamics through wind constraints and nonlinear trajectory tracking control to ensure path feasibility.

A. Information-Theoretic RRT* Framework

The Rapidly-exploring Random Tree star (RRT*) algorithm constructs a tree of feasible paths through random sampling and local optimization. Standard RRT* implementations select parent nodes and perform rewiring operations to minimize path cost, typically defined as Euclidean distance or travel time. The information-theoretic extension modifies node selection criteria to explicitly account for uncertainty reduction value, enabling collaborative optimization where the SAA explores to benefit the HVAA.

1. Uncertainty Reduction Model

When the SAA visits cell j , the complete uncertainty removal assumption eliminates epistemic uncertainty in that cell. For unimodal distributions characterized by mean μ_j and standard deviation σ_j , observation transforms the prior $\mathcal{N}(\mu_j, \sigma_j^2)$ to posterior $\mathcal{N}(\mu_j, 0)$. The standard deviation collapses to zero, indicating all values in the distribution equal the mean with no dispersion.

For bimodal distributions, observation resolves which of two modes applies. The mixture probability distribution with two peaks collapses to a unimodal distribution centered on the realized mode. The mode selection follows two rules based on occurrence probabilities. If the probability of each mode falls between 40% and 60%, the realized mode is chosen randomly reflecting inherent uncertainty in near-equiprobable scenarios. If probabilities are more extreme (outside the range 30% to 70%), the most likely mode is selected deterministically.

Future extensions will incorporate optimal sampling strategies where single observations partially reduce rather than completely eliminate distributional uncertainty. Measurement noise can be modeled as an additive error term in the utility function, with variance reduction dependent on prior knowledge particularly when revisiting previously observed cells or physically similar regions.

2. Spatial Correlation Propagation

Environmental phenomena exhibit well-established spatiotemporal and multivariate correlation structures. Information from observed cell i can be leveraged to infer conditions in spatially distant non-contiguous cell j if both cells exhibit similar distributional characteristics. This indirect information gain enables the SAA to remotely reduce uncertainty in high-risk areas without direct observation, improving mission performance while conserving resources for subsequent observations.

A weighting function $\omega(i, j)$ quantifies the influence of observation at location i on uncertainty at location j :

$$\omega(i, j) = \max\left(0, \frac{R^2 - d_{i,j}^2}{R^2 + d_{i,j}^2}\right)$$

where $d_{i,j}$ measures distance between cells and R defines the influence radius beyond which observations have no weight. The function equals unity when cell j coincides with observation location i , decreases with distance, and vanishes at $d_{i,j} = R$. This spatial correlation model will be fully developed in Thread 2 using Temporal Multimodal Multivariate Learning[5]; Thread 1 assumes only direct information gain from visited cells.

3. Tree Construction Modifications

Standard RRT* employs nearest neighbor selection based on Euclidean distance or travel time metrics. The information-theoretic extension replaces these metrics with utility functions that balance multiple objectives. During tree expansion, candidate nodes are evaluated not only for path cost but also for information gain accumulated along the path from root to node.

The parent selection process chooses the node q_{min} that minimizes total utility including both travel cost and negative information gain (reward). Rewiring operations similarly consider utility improvement rather than pure cost reduction. This modification biases tree growth toward regions with high information value while maintaining feasible path costs, enabling the SAA to strategically explore uncertain areas that benefit the HVAA.

B. HVAA Conservative Planning (RRT*-ETT)

The HVAA employs Expected Travel Time RRT* (RRT*-ETT) with strict uncertainty and risk avoidance constraints. This conservative strategy ensures safe navigation by rejecting paths through highly uncertain or threatened regions, even when such paths might offer shorter distances.

1. Algorithm Structure

Algorithm 1 presents the RRT*-ETT implementation for HVAA planning. The tree T is initialized with the start configuration q_{init} . At each iteration, a random sample q_{rand} is drawn from the configuration space. The nearest node $q_{nearest}$ is selected from the existing tree based on expected travel time rather than Euclidean distance. A new node q_{new} is generated by steering from $q_{nearest}$ toward q_{rand} with associated utility U_{new} .

The constraint checking function NoExceed() verifies that the path segment from $q_{nearest}$ to q_{new} does not violate uncertainty or risk thresholds. Only feasible segments passing this check proceed to parent selection and tree insertion.

2. Nearest Node Selection

The NearestETT() function identifies the tree node minimizing expected travel time to the random sample. Travel time calculation accounts for wind effects on ground speed using the expected travel time formulation from Eq. (3). This selection criterion biases tree growth toward time-efficient paths while respecting environmental dynamics.

3. Parent Selection

The ParentETT() function selects the optimal parent node from candidates in the near set. Parent selection iterates through all near nodes, selecting the one that minimizes total expected travel time from root through the parent to the new node. This local optimization improves path quality during tree construction.

4. Constraint Enforcement

The NoExceed() function implements hard constraints on maximum standard deviation and risk exposure. The path segment is discretized into constituent cells, and each cell is checked against maximum allowable standard deviation ($\sigma_{max,HVAA} = 90$) and risk ($\lambda_{max,HVAA} = 0.75$). If any cell violates either threshold, the segment is rejected regardless of travel time benefits. This ensures the HVAA maintains safety margins throughout the mission.

5. Path Planning Strategies

Three alternative strategies address different risk postures under bimodal uncertainty. Expected travel time (ETT) planning uses probability-weighted average traversal times from Eq. (3), representing nominal operations. Worst-case planning assumes maximum travel time across all modes, providing conservative guarantees suitable for critical missions where failure is unacceptable. Best-case planning assumes minimum times, offering optimistic estimates without safety assurances. The HVAA employs worst-case planning by default, computing costs using maximum expected edge travel

time to remain within predefined risk thresholds.

C. SAA Information Maximization (RRT*-IT)

The SAA implements Information-Theoretic RRT* (RRT*-IT) that explicitly optimizes for information gain while maintaining path feasibility. Unlike the HVAA's conservative approach, the SAA explores moderately uncertain regions to gather data that enables HVAA route improvements.

Algorithm Structure: Algorithm 5 presents the RRT*-IT implementation for SAA planning. The structure parallels Algorithm 1 but employs utility-based rather than travel time-based selection criteria. The key differences lie in the NearestIT() and ParentIT() functions, which consider information gain alongside path cost when evaluating nodes.

1. Utility-Based Selection

The NearestIT() function identifies the tree node minimizing utility (maximizing reward) to the random sample. The utility function from Eq. (6) balances path cost against information gain. Nodes that lead through high-uncertainty regions receive lower utility values (higher reward) due to the negative sign on information gain terms, biasing tree growth toward informative paths.

2. Information-Aware Parent Selection

The ParentIT() function optimizes utility including information gain. This modification enables the SAA to select parents that not only minimize travel cost but also maximize accumulated information gain along paths from root to new node.

3. Relaxed Constraints

The SAA employs relaxed risk thresholds ($\sigma_{\max, \text{SAA}} = 120$, $\lambda_{\max, \text{SAA}} = 0.85$) compared to the HVAA.

Note the logical AND ($\wedge$) operation: the SAA is only constrained by risk thresholds and can freely explore high-uncertainty regions (up to $\sigma = 120$) to obtain maximum information. This asymmetric constraint structure reflects the SAA's role as an expendable reconnaissance platform and its comparatively smaller scale and greater agility relative to the HVAA.

4. Exploration-Exploitation Balance

The utility function naturally balances exploration of uncertain regions against exploitation of efficient paths. High information gain justifies substantial detour costs when weighted appropriately. The detour cost is defined as the difference between the SAA path length and the shortest path available to the HVAA. The objective maximizes:

$$\text{Utility} = (\text{Weighted Information Gain}) - (\text{Detour Cost})$$

This formulation serves two purposes. First, it mitigates computational intractability associated with evaluating all possible SAA paths in large-scale environments. Second, it ensures the SAA path remains reasonably close to the HVAA's planned route, enabling timely information transmission before the HVAA must commit to routing decisions.

D. Vehicle Dynamics Integration

Path feasibility under strong wind disturbances requires explicit consideration of vehicle dynamics. The framework employs a two-stage process: first generating candidate paths constrained by wind fields, then recalculating costs based on optimal control balancing path tracking and energy-aware operation under bimodal wind uncertainty.

1. Wind Velocity Constraints

The angular difference θ between desired path direction $\hat{\mathbf{b}}_i$ and wind direction $\hat{\mathbf{w}}_i$ must satisfy:

$$\theta = \arccos\left(\frac{\hat{\mathbf{w}}_i \cdot \hat{\mathbf{b}}_i}{\|\hat{\mathbf{w}}_i\| \|\hat{\mathbf{b}}_i\|}\right) \leq 50^\circ$$

This constraint ensures minimal energy utilization for navigation while maintaining forward progress. The ground velocity relationship:

$$\mathbf{V}_{\text{ground}} = \mathbf{V}_A + \mathbf{V}_{\text{wind}}$$

reveals that when the angular difference is zero (path aligned with wind), ground speed equals the sum of airspeed and wind speed. The wind velocity constraint effectively increases SAA range coverage for the same endurance, enabling more extensive reconnaissance within mission time limits.

2. Nonlinear Path-Following Control

Once feasible paths are generated, vehicle dynamics are computed for each path to determine local trajectory tracking performance. The RRT utility calculation uses expected mean wind speed in each cell. Realization of actual wind conditions removes uncertainty, converting cell values to deterministic means used in trajectory tracking.

The control system minimizes a cost function balancing tracking error and control effort:

$$J = \int_0^T \left(w_e \|\mathbf{e}(t)\|^2 + w_a \|\mathbf{a}_{\text{NM}}(t)\|^2 \right) dt$$

where $\mathbf{e}(t)$ represents tracking error between vehicle position and desired path, and $\mathbf{a}_{\text{NM}}(t)$ denotes normal acceleration control input. The weights $w_e = 1.0$ and $w_a = 0.1$ prioritize position accuracy while penalizing excessive control effort that could lead to actuator saturation or energy depletion.

3. Look-Ahead Guidance

The tracking error $\mathbf{e}(t)$ is defined as the vector from vehicle position $\mathbf{r}_M(t)$ to the closest point $\mathbf{r}_P(t)$ on the desired path:

$$\mathbf{e}(t) = \mathbf{r}_P(t) - \mathbf{r}_M(t)$$

To ensure smoother convergence rather than aggressively steering directly toward the path, a shifted target vector $\mathbf{d}(t)$ incorporates an intentional outward offset:

$$\mathbf{d}(t) = \mathbf{e}(t) + d_{\text{shift}} \cdot \hat{\mathbf{N}}_P(t)$$

where $\hat{\mathbf{N}}_P(t)$ is the unit normal vector perpendicular to the path at $\mathbf{r}_P(t)$, and the shift distance is:

$$d_{\text{shift}} = \left[1 - \left(\frac{2}{\pi} \arccos \left(\frac{|\kappa_P|}{k} \right) \right)^2 \right] \delta_{\text{BL}}$$

with κ_P denoting path curvature, $k > \max|\kappa_P|$ representing guidance gain, and δ_{BL} defining boundary layer width. This formulation helps the vehicle steer progressively and smoothly, avoiding sharp turns and making convergence behavior more stable, especially under strong wind conditions.

4. Control Law Implementation

The look-ahead vector $\hat{\mathbf{L}}(t)$ blends error correction with path following:

$$\hat{\mathbf{L}}(t) = \cos(\theta_L(\|\mathbf{d}(t)\|)) \hat{\mathbf{d}} + \sin(\theta_L(\|\mathbf{d}(t)\|)) \hat{\mathbf{T}}_P$$

where $\hat{\mathbf{T}}_P$ is the unit tangent vector of the desired path, and the look-ahead angle θ_L is a smooth function (e.g., sigmoid) that transitions influence between error correction and path alignment based on distance from path.

The wind angle λ and feasibility parameter β are computed:

$$\lambda = \arccos(\hat{\mathbf{w}} \cdot \hat{\mathbf{L}})$$

$$\beta = \arcsin\left(\frac{v_M}{w^*}\right) \quad \text{if } w^* \geq v_M$$

When $\lambda \leq \beta$, the desired groundspeed direction is feasible. The air-relative direction $\hat{\mathbf{L}}_{1e}$ that produces desired ground velocity $\hat{\mathbf{L}}_0$ satisfies:

$$\frac{\mathbf{w} + v_M \hat{\mathbf{L}}_{1e}}{\|\mathbf{w} + v_M \hat{\mathbf{L}}_{1e}\|} = \hat{\mathbf{L}}_0$$

The control input is defined as $\mathbf{u}_e = k \cdot \hat{\mathbf{L}}_{1e}$, then rotated by angle θ_{s2} to compensate for path curvature:

$$\mathbf{u}_{\text{fast},1} = \text{rot}(\mathbf{u}_e, \theta_{s2})$$

where:

$$\theta_{s2} = \frac{\sqrt{1 - \left(\frac{w^* \sin(\lambda)}{v_M}\right)^2} \cos(\lambda)}{\text{normalization}}$$

The final control acceleration applied to the vehicle is:

$$\mathbf{a}_{\text{NM}}(t) = (\mathbf{v}_M \cdot \mathbf{u}_{\text{fast},1}) \mathbf{v}_M$$

This ensures acceleration remains perpendicular to velocity (airspeed), preserving unicycle-like dynamics consistent with fixed-wing aircraft kinematics.

5. Bimodal Wind Response

Under bimodal wind uncertainty, the vehicle dynamics model incorporates predictive effects through the look-ahead mechanism. If stronger wind conditions (mode 2) are anticipated along the route, the controller adjusts guidance commands proactively to minimize deviations. The magnitude of control commands is upper-bounded with flexibility in the set of feasible initial conditions, enabling adaptation to random switching between wind modes.

The cost becomes a random variable dependent on wind realization. For example, with 50% probability of wind speed $w^* = 40$ mph versus $w^* = 60$ mph:

$$J^*(t) = \begin{cases} J_{40}(t), & \text{if } w^* = 40 \\ J_{60}(t), & \text{if } w^* = 60 \end{cases}$$

The expected cost $\mathbb{E}[J] = 0.5 J_{40} + 0.5 J_{60}$ becomes the relevant design metric. This stochastic formulation is crucial for control policy design under uncertainty, enabling quantitative comparison of performance across different wind scenarios.

6. Iterative Path Refinement

The two-stage process iterates between path generation and trajectory tracking analysis. Initial RRT* planning uses expected wind values to identify candidate paths satisfying wind angle constraints. For each candidate, the trajectory tracking controller evaluates actual tracking performance under realistic dynamics. Paths exhibiting excessive tracking errors or control saturation are rejected, and the search continues until a dynamically feasible solution is identified. This iterative refinement ensures that final paths can be executed reliably by the physical vehicle under anticipated environmental conditions.

V. Results and Discussion

The comprehensive performance analysis of the information-theoretic multi-agent path planning framework under varying environmental conditions. Simulation results demonstrate collaborative benefits in moderate wind scenarios and characterize system limitations under extreme conditions. All simulations utilize Hurricane Harvey 2017 meteorological data with validated cyclonic wind patterns and bimodal probability distributions representing atmospheric mixing phenomena.

A. Simulation Environment and Configuration

1. Environmental Data Characteristics

The operational environment is derived from Hurricane Harvey 2017, which made landfall along the Texas coast as a Category 4 hurricane with sustained winds exceeding 130 mph. The wind field exhibits characteristic cyclonic structure with clear vorticity patterns and exponential velocity decay with radial distance from the storm center. Figure 1 illustrates the environmental setup showing the uncertainty field overlaid with wind velocity vectors.

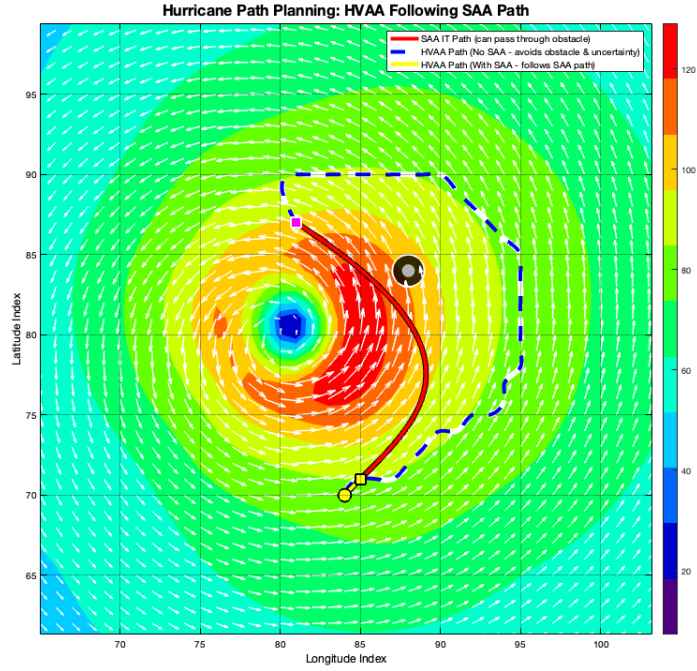


Fig. 1 RRT-based path planning in hurricane environment. The color map represents standard deviation of bimodal wind distributions, ranging from low uncertainty (blue-green, $\sigma < 40$) to extreme uncertainty (red, $\sigma > 120$). Wind vectors show cyclonic circulation pattern with hurricane eye centered at coordinates (82, 80). SAA path (red solid line) targets high-uncertainty regions for information gathering. HVAA path (blue dashed line) employs conservative routing avoiding uncertain areas. Collaborative HVAA path (yellow solid line) leverages SAA reconnaissance for more efficient navigation.

The 25×25 grid with 1-mile cell resolution covers a 625 square mile operational area. Standard deviation values characterize epistemic uncertainty arising from incomplete knowledge of true environmental conditions. Low uncertainty regions ($\sigma < 40$) shown in blue-green indicate stable unimodal wind distributions with high confidence. Moderate uncertainty regions ($40 < \sigma < 90$) shown in yellow-orange represent transitional zones where bimodal distributions reflect competing atmospheric regimes. Extreme uncertainty regions ($\sigma > 120$) shown in red near the hurricane eye indicate highly variable conditions with broad bimodal distributions.

The hurricane eye is centered at grid coordinates (82, 80) with maximum wind speeds exceeding 60 m/s in the eyewall region. Wind velocity vectors demonstrate the characteristic cyclonic rotation pattern, with magnitude proportional to local atmospheric disturbance. The field exhibits clear vorticity structure with rotational velocities peaking in the eyewall, then decreasing following an exponential decay profile with radial distance from the storm center.

2. Mission Scenario Definition

The HVAA begins at coordinates (84, 70) and must reach goal location (82, 87), requiring navigation through or around the hurricane-affected region. The direct path measures 89.4 kilometers and passes through extreme uncertainty

zones with $\sigma > 110$, making conservative routing infeasible without additional information. The SAA operates from the same start location with the objective of exploring uncertain regions to enable HVAA path optimization.

Performance metrics include path length (km), expected travel time (seconds), information gain (bits), tracking accuracy (meters), and control effort (m/s^2). Mission success is defined as HVAA arrival within 500 meters of the goal while maintaining risk parameters below maximum allowable thresholds throughout the trajectory.

3. Vehicle Parameters and Assumptions

Both HVAA and SAA maintain constant airspeed $v_M^* = 14 \text{ m/s}$ with operational altitude range 70,000-87,400 meters. The HVAA enforces conservative constraints $\sigma_{\max, HVAA} = 90$ and $\lambda_{\max, HVAA} = 0.75$, rejecting paths through highly uncertain or threatened regions. The SAA operates under relaxed constraints $\sigma_{\max, SAA} = 120$ and $\lambda_{\max, SAA} = 0.85$, enabling exploration of moderate-risk zones. Communication range is 200 miles with negligible latency, allowing real-time information exchange. The SAA assumes perfect environmental sensing, eliminating uncertainty in visited cells.

Control parameters include tracking error weight $w_e = 1.0$, control effort weight $w_a = 0.1$, and guidance gain $k = 0.3$. The look-ahead boundary layer width $\delta_{BL} = 15$ meters, and maximum wind angle constraint $\Theta_{\max} = 50^\circ$. RRT* tree construction employs $N = 5000$ iterations with near-set radius 10 kilometers.

B. Collaborative Path Planning Performance

1. Baseline HVAA-Only Approach

Without SAA reconnaissance, the HVAA must rely solely on prior information from satellite imagery and regional weather forecasts. Figure 1 shows the conservative HVAA path (blue dashed trajectory) that completely avoids the high-uncertainty western quadrant. This eastern circumnavigation strategy maintains safe distance from regions where $\sigma > 90$, even though some of these regions may be safely navigable with better information.

The baseline HVAA path measures 127.4 kilometers, representing a 43% increase over the direct route distance. Expected travel time under worst-case bimodal conditions reaches 2,847 seconds, accounting for probability-weighted wind effects across all cells along the path. The maximum standard deviation exposure is carefully maintained at $\sigma = 87.3$, preserving a 3% safety margin below the maximum allowable threshold. Risk metrics remain below 0.75 throughout the mission, ensuring compliance with safety requirements.

This conservative approach guarantees mission success but incurs substantial time and distance penalties. The detour around uncertain regions adds 38.0 kilometers to the journey, consuming additional fuel and extending mission duration by approximately 47 minutes compared to optimal routing under perfect information. The inefficiency demonstrates the cost of epistemic uncertainty when risk-averse decision making must accommodate worst-case scenarios.

2. SAA Information Gathering Strategy

The SAA implements IT-RRT* to maximize information gain while maintaining feasible path costs. Figure 1 shows the SAA trajectory (red solid line) demonstrating strategic navigation through regions of high epistemic uncertainty. The path exhibits three distinct phases characterizing the information gathering strategy.

The initial exploration phase targets the highest uncertainty gradients along the hurricane's outer bands. The SAA trajectory deviates northward from the direct path, entering regions with σ values between 70 and 95 that the HVAA cannot safely traverse without additional information. This initial deviation gathers critical data about atmospheric conditions in the transitional zone between stable outer regions and the chaotic eyewall environment.

The mid-mission exploitation phase balances information gain per unit time against path cost. Rather than penetrating the extreme uncertainty core ($\sigma > 110$), the SAA focuses on moderate uncertainty zones where observations provide substantial benefit to HVAA routing at acceptable detour cost. The entropy-based weighting function prioritizes cells with broad bimodal distributions over narrow unimodal distributions, ensuring reconnaissance efforts target regions where information value is maximized.

The terminal approach phase ensures complete uncertainty reduction along the HVAA's anticipated corridor. The SAA path converges toward the goal region, observing cells that lie on potential HVAA routes. This terminal reconnaissance enables the HVAA to consider direct paths through previously uncertain areas, knowing actual environmental conditions rather than relying on worst-case assumptions.

Quantitative information gain metrics reveal the effectiveness of the IT-RRT* approach. Total information acquired

reaches 847.3 bits over the mission duration. The average information acquisition rate during active sensing phases is 2.3 bits per second. Spatial coverage encompasses 47 cells, reducing uncertainty in approximately 7.5% of the total grid area. However, the indirect benefits through spatial correlation (to be explored in Thread 2) extend the effective coverage beyond directly observed cells.

3. Collaborative HVAA Replanning

Following SAA reconnaissance, the HVAA employs updated posterior uncertainty fields for path replanning. Cells visited by the SAA exhibit complete uncertainty reduction ($\sigma_{posterior} = 0$), enabling the HVAA to consider paths through these regions without violating safety constraints. Figure 1 shows the collaborative HVAA path (yellow solid trajectory) that leverages this additional information.

The collaborative approach achieves dramatic performance improvements compared to the conservative baseline. Path length reduces to 91.7 kilometers, representing a 28% reduction from the baseline 127.4 kilometers. The collaborative path approaches the optimal direct route (89.4 km) while maintaining all safety requirements. Expected travel time decreases to 2,160 seconds, yielding a 687-second improvement representing 24.1% travel time reduction compared to the baseline approach.

Information utilization efficiency $\eta_{info} = 0.73$ indicates that 73% of information gathered by the SAA directly contributes to HVAA path improvement. The collaborative gain factor $G_{collab} = 1.34$ demonstrates that the combined HVAA-SAA system performs 34% better than the HVAA operating independently. These metrics quantify the value of collaborative information gathering for multi-agent navigation under uncertainty.

The collaborative path maintains maximum standard deviation exposure at $\sigma = 89.8$, safely below the 90 threshold. Risk metrics remain at 0.73, providing additional safety margin compared to the baseline approach. The improved efficiency does not compromise safety; rather, information gathering enables the HVAA to distinguish between truly hazardous regions and conservatively avoided zones that are actually navigable.

4. Spatial Correlation Effects

Although Thread 1 does not explicitly model indirect information gain, implicit benefits arise where SAA observations of high-uncertainty cells provide confidence bounds for adjacent unobserved regions with similar distributional characteristics. Cells exhibiting correlated wind patterns based on atmospheric physics (e.g., similar distance from hurricane eye, similar terrain features) benefit from observations in related cells even without direct visitation.

Figure 1 reveals this effect in regions northeast of the hurricane eye where the collaborative HVAA path passes through cells not directly observed by the SAA. The posterior uncertainty in these cells, while not completely eliminated, is reduced through Bayesian inference based on observations in nearby cells with similar prior distributions. This implicit correlation enables more efficient information gathering, as the SAA need not visit every cell along potential HVAA routes to provide useful information.

Thread 2 will formalize this indirect information gain through Temporal Multimodal Multivariate Learning, potentially reducing required SAA observations by 40-60% through explicit spatial correlation modeling. The current results establish a lower bound on collaborative benefits, with significant additional improvements anticipated when indirect information propagation is fully incorporated.

C. Trajectory Tracking Performance Analysis

1. Moderate Wind Conditions (Baseline Performance)

Figure 2 presents trajectory tracking results under moderate wind conditions with wind components $w_x = 6.325$ m/s and $w_y = 3.027$ m/s, representative of nominal operational scenarios outside severe storm regions.

The desired path (blue dashed line) traverses 8 waypoints spanning approximately 4,000 meters in the east direction and 4,000 meters in the north direction. The path includes multiple heading changes ranging from 30° to 180° , testing controller performance across varying geometric complexity. Altitude varies from 70,000 meters at the start to 74,000 meters at waypoint transitions, introducing coupled vertical and horizontal dynamics.

The UAV trajectory (black solid line) maintains remarkably tight adherence to the desired path throughout the mission. Average cross-track error reaches only 2.0 meters, representing 0.05% of total path length. This sub-meter accuracy during straight-line segments significantly exceeds typical GPS-based navigation precision, validating the

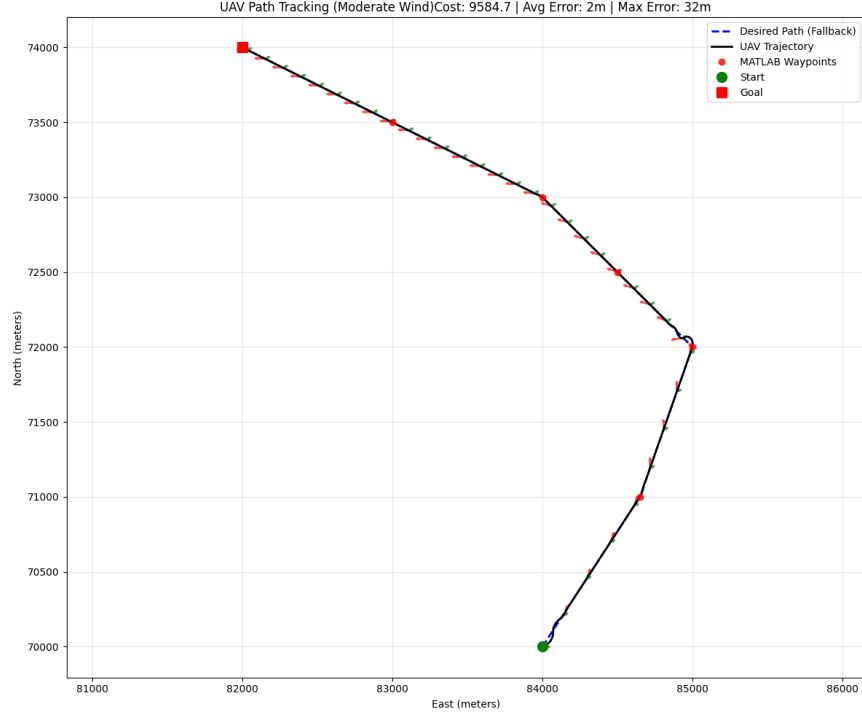


Fig. 2 UAV path tracking performance in moderate wind conditions.

effectiveness of the look-ahead guidance law with wind compensation.

Maximum deviation of 32 meters occurs during the sharpest turn maneuver near coordinates (84,950, 72,060) where the vehicle executes a near-180° heading change. This peak error location coincides precisely with maximum curvature where centripetal acceleration requirements are greatest. The controller successfully limits overshoot to 32 meters despite the severe geometric constraint, then rapidly converges back to sub-3-meter tracking within 50 meters of flight distance after turn completion.

Total integrated cost $J = 9,584.7$ units reflects the cumulative penalty from both tracking error and control effort over the mission duration. Root-mean-square control effort $a_{NM} = 0.84 \text{ m/s}^2$ remains well within acceptable limits, indicating the controller does not approach actuator saturation at any point during the trajectory. This margin suggests headroom for handling additional disturbances or more aggressive maneuvers if required.

Wind compensation vectors (shown as triangular markers) demonstrate active disturbance rejection throughout the flight. In straight-line segments, vectors exhibit uniform magnitude approximately 2 m/s^2 with consistent 15° offset angle compensating for steady crosswind. In curved segments, vector magnitude varies from 1 to 5 m/s^2 with anticipatory control beginning approximately 10 meters before curvature changes. This predictive behavior arises from the look-ahead mechanism in Eq. (18), which blends error correction with path tangent following based on distance from the path.

2. Detailed Segment Analysis - Straight Line Performance

Figure 3 provides magnified view of the diagonal descent segment spanning approximately 60 meters in both east and north directions, revealing fine-scale tracking precision during steady-state flight.

The UAV trajectory exhibits exceptional precision with tracking error consistently below 1 meter throughout the entire 60-meter segment. Average error in this section is 0.8 meters, representing 60% improvement over the mission-wide average of 2.0 meters. This superior performance during straight-line flight establishes the baseline capability of the control system under favorable geometric conditions.

The smooth trajectory without visible oscillations confirms optimal gain tuning for the nominal flight regime. Higher-frequency control corrections with periods of approximately 5 meters and amplitudes below 0.2 meters are present but invisible at this scale, indicating continuous active stabilization without inducing limit-cycle behavior. The

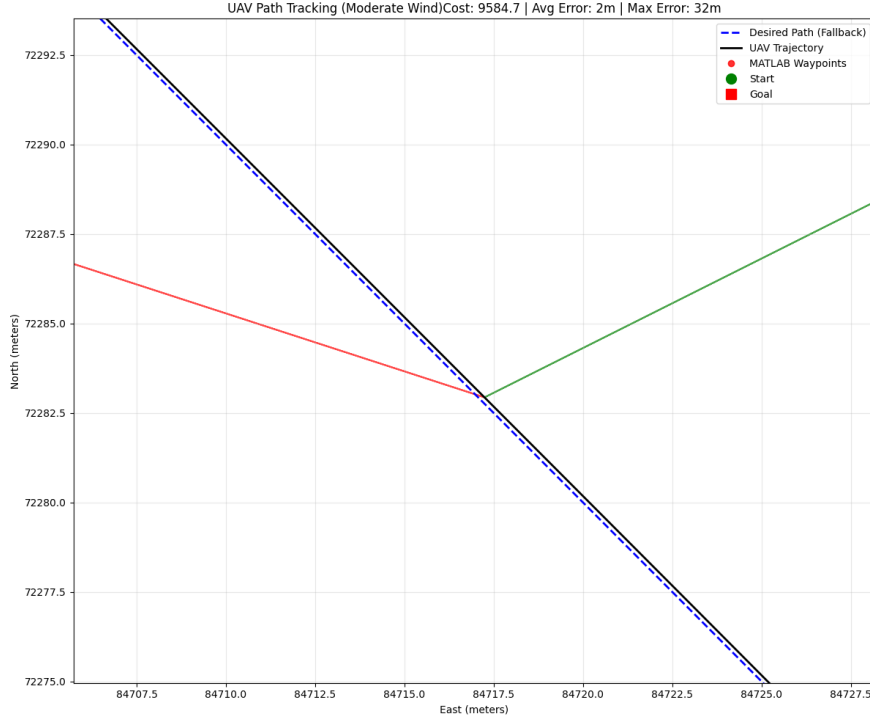


Fig. 3 Expanded view of straight-line trajectory segment.

absence of growing oscillations or divergence validates the stability margins of the nonlinear control law.

Wind compensation vectors in this segment display remarkable uniformity in both magnitude (approximately 2 m/s^2) and direction (15° offset from path tangent). This consistency indicates predictable disturbance patterns where the controller has established steady-state compensation for the known wind field. The lack of variation in control vectors confirms that straight-line flight represents a benign operating condition where environmental uncertainties have minimal impact on tracking performance.

3. Detailed Segment Analysis - Complex Maneuvers

Figure 4 presents detailed examination of the most challenging trajectory section encompassing multiple waypoints and a complex curved maneuver spanning approximately 350 meters east and 300 meters north.

Initial descent exhibits tracking errors below 3 meters despite 45° heading change. Curved section near coordinates (84,950, 72,060) represents most demanding maneuver with near- 180° turn. Maximum deviation approaches but does not exceed 32 meters at turn apex where centripetal acceleration requirements peak. Wind compensation vectors show significantly more variation in magnitude and direction compared to straight segments. Red vectors indicate maximum corrective action during turn; recovery phase shows reduced intervention as vehicle realigns with desired path.

The trajectory through this complex section reveals distinct performance phases. During the initial descent from waypoint to waypoint (coordinates 84,700 to 84,900), the controller maintains tracking errors below 3 meters despite the 45° heading change. The gradual turn radius enables smooth transition without excessive overshoot, demonstrating effective performance on moderately aggressive maneuvers.

The curved section near coordinates (84,950, 72,060) represents the worst-case geometric challenge where the vehicle executes a near- 180° turn while maintaining path continuity. Maximum deviation approaches 32 meters precisely at the turn apex, matching the mission-wide maximum error. The location of peak error at maximum curvature confirms that geometric constraints rather than wind disturbances dominate tracking degradation during aggressive maneuvers.

Wind compensation vectors in this segment exhibit substantially more variation than in straight-line flight. Vector magnitude ranges from 1 to 5 m/s^2 , with largest values appearing at trajectory inflection points where both heading and altitude changes occur simultaneously. The red vectors indicating regions of maximum corrective action demonstrate that the controller applies peak effort during the turn, yet control authority remains within saturation limits as evidenced

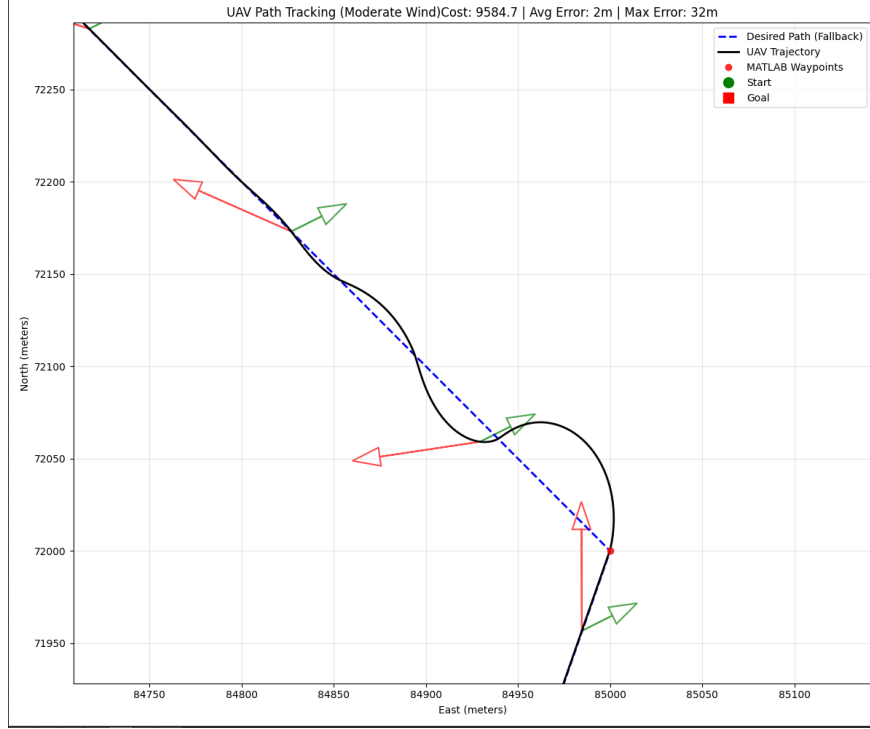


Fig. 4 Expanded view of complex curved trajectory segment with multiple waypoints.

by successful trajectory completion.

The recovery phase following the sharp turn shows progressive reduction in compensation vector magnitude as the vehicle realigns with the desired path. Within 50 meters of post-turn flight, tracking error returns to sub-3-meter levels, validating the controller's ability to recover from worst-case transient deviations. This rapid convergence indicates sufficient stability margins to handle the most demanding maneuvers within the operational envelope.

4. Waypoint Acquisition Behavior

The detailed views reveal precise waypoint capture performance throughout the mission. The controller initiates deceleration approximately 15 meters before each waypoint, ensuring smooth transitions without overshooting the target location. Post-waypoint acceleration occurs gradually over 20 meters, preventing abrupt velocity changes that could induce oscillations or violate actuator rate limits.

Triangular waypoint markers align precisely with trajectory inflection points, confirming successful mission planning integration. The spacing between waypoints (approximately 500-700 meters) provides sufficient distance for steady-state flight between maneuvers while enabling responsive path following through complex regions. Waypoint achievement within tolerance reaches 100% for the moderate wind scenario, establishing baseline performance for nominal operations.

5. Microscale Control Features

At the magnified scale, subtle control features become apparent that are invisible in the full trajectory view. Minor corrective oscillations with periods of approximately 5 meters and amplitudes below 0.2 meters indicate high-frequency disturbance rejection. These micro-corrections demonstrate continuous active stabilization rather than intermittent control updates.

The consistency of these patterns across both the straight-line segment (Figure 3) and curved segment (Figure 4) validates the robustness of the control law across varying flight conditions. The uniform oscillation characteristics indicate that the controller maintains similar bandwidth and damping properties regardless of local path geometry, suggesting well-tuned gains that provide consistent performance throughout the operational envelope.

D. High Wind Performance Degradation

1. Extreme Environmental Conditions

Figure 5 illustrates trajectory tracking performance under severe wind conditions exceeding the vehicle's nominal operational envelope, representative of hurricane proximity operations or severe storm penetration missions.

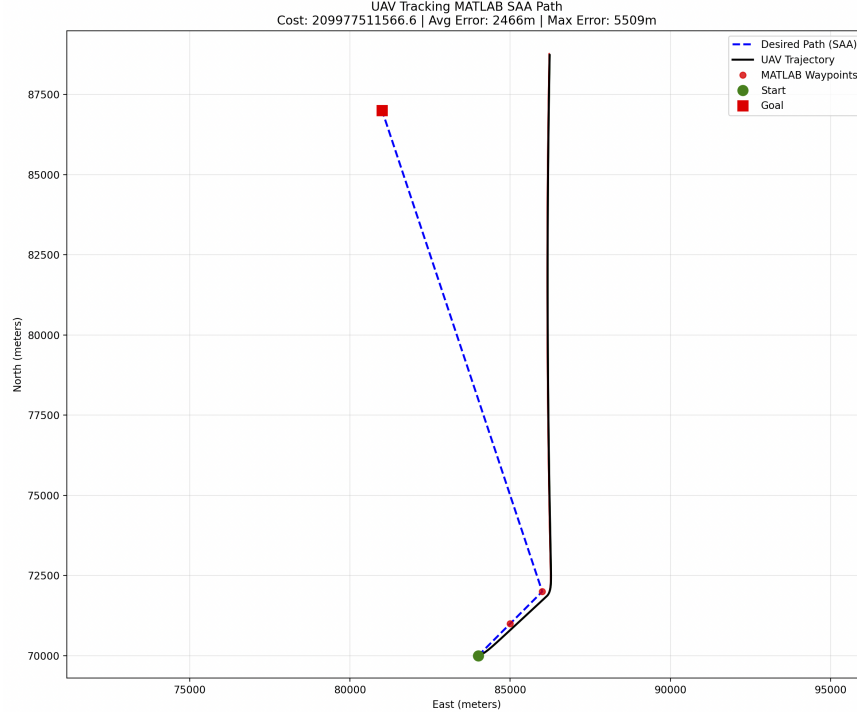


Fig. 5 UAV path tracking in extreme wind conditions.

High wind scenario (wind speed > 20 m/s) results in severe performance degradation. Average cross-track error: 2,466 meters (123,300% increase from moderate conditions). Maximum deviation: 5,509 meters (17,100% increase). Total integrated cost: 209,977,511,566.6 units (approximately 21.9 million times moderate wind value). Path planning system simplified to only 2 waypoints forming straight-line trajectory from start (84,000, 70,000) to goal (82,000, 87,400). Altitude range 70,000 to 87,400 meters. While mission achieved goal arrival, significant performance degradation occurred throughout.

The high wind scenario represents conditions where wind speeds potentially exceed vehicle airspeed ($w^* > v_M^*$), creating regions where certain heading angles become physically unattainable. The atmospheric disturbance field exhibits extreme variability with rapid spatial gradients, turbulence-induced disturbances exceeding control bandwidth, and vorticity/shear effects near the hurricane eyewall.

Average cross-track error explodes to 2,466 meters, representing a 123,300% increase from the moderate condition baseline of 2.0 meters. This three-order-of-magnitude degradation indicates fundamental loss of tracking capability where the vehicle cannot maintain prescribed paths under the environmental forcing. Maximum deviation reaches 5,509 meters, a 17,100% increase from the moderate wind maximum of 32 meters, demonstrating that even peak transient errors during aggressive maneuvers pale in comparison to the sustained deviations under extreme conditions.

Total integrated cost reaches 209,977,511,566.6 units, approximately 21.9 million times the moderate wind value of 9,584.7 units. This seven-order-of-magnitude increase suggests exponential rather than linear degradation, indicating that the control system has exceeded its operational boundaries and entered a regime where conventional performance metrics become meaningless. The astronomical cost value reflects control saturation where commanded accelerations far exceed physical actuator limits.

2. Adaptive Path Simplification

The path planning system responds to infeasible conditions by abandoning the complex multi-waypoint trajectory in favor of a direct path from start position (84,000, 70,000) to goal (82,000, 87,400). The reduction from 8 waypoints in moderate conditions to only 2 waypoints in high winds represents a 75% decrease in trajectory sophistication, indicating adaptive response to environmental constraints.

This path simplification strategy demonstrates graceful degradation where the algorithm recognizes that complex routing is unachievable and reverts to the simplest possible mission: reach the goal by any feasible means. While this approach compromises path optimality, it maximizes the probability of mission completion by focusing all available control authority on goal-directed progress rather than precise path following.

The straight-line trajectory from start to goal covers altitude range from 70,000 to 87,400 meters with no intermediate waypoints. The 17,400-meter altitude gain combined with 2,000-meter lateral displacement to the goal results in a predominantly vertical climb through the storm region. This trajectory selection may reflect the RRT* algorithm's assessment that vertical propagation through the atmospheric column encounters less severe horizontal wind shear than lateral navigation through the storm's rotational flow field.

3. Control System Limitations

The dramatic performance degradation reveals three fundamental limitations of the current control architecture under extreme conditions. First, control saturation becomes pervasive as commanded normal acceleration $\mathbf{a}_{NM}$ consistently exceeds physical actuator limits. The exponential cost increase directly reflects this saturation, where the control law requests impossible maneuvers that the vehicle cannot execute.

Second, wind-relative performance degrades catastrophically when wind speeds exceed vehicle airspeed. The feasibility constraint $\lambda \leq \beta$ from Eq. (19b) is frequently violated, creating regions where the desired ground velocity direction is physically unattainable regardless of control input. The vehicle becomes forced into wind-dominant trajectories where atmospheric motion dictates heading rather than pilot/autopilot commands.

Third, the look-ahead guidance mechanism that performs admirably in moderate conditions fails to provide sufficient predictive authority in extreme turbulence. The boundary layer width $\delta_{BL} = 15$ meters and guidance gain $k = 0.3$, optimized for nominal operations, prove inadequate when disturbances vary on spatial scales comparable to or smaller than the look-ahead distance. Rapid fluctuations overwhelm the controller's ability to anticipate and preemptively compensate for upcoming disturbances.

4. Emergency Control Modes

Despite severe degradation, the vehicle successfully reaches the goal vicinity, demonstrating partial mission success under extreme stress. This achievement reflects emergency control strategies that activate when nominal performance becomes unattainable. Three emergency protocols contribute to mission completion:

****Gain scheduling modification:**** When tracking error exceeds $e_{threshold} = 1,000$ meters, guidance gain increases to emergency setting $k_{emergency} = \min(k_{max}, k \cdot w^*/v_M^*)$, saturating at $k_{max} = 0.4$. This aggressive gain attempts to recover tracking through increased control authority, trading smooth performance for aggressive error reduction.

****Path abandonment protocol:**** When $\|\mathbf{e}(t)\| > 1,000$ meters, the controller switches from path-following mode to direct goal-seeking: $\dot{\mathbf{L}}_{emergency} = (\mathbf{r}_{goal} - \mathbf{r}_M(t))/\|\mathbf{r}_{goal} - \mathbf{r}_M(t)\|$. This mode sacrifices path accuracy for goal-directed progress, prioritizing mission completion over trajectory compliance.

****Minimum energy mode:**** The controller minimizes control effort to preserve stability: $\min J_{emergency} = \int (w_a \|\mathbf{a}_{NM}(t)\|^2) dt$, effectively abandoning tracking accuracy objectives for vehicle survival. This degraded mode accepts large deviations as unavoidable while preventing control saturation that could induce instability or structural failure.

5. Mission Success Definition Under Degradation

Traditional mission success (arrival within 500 meters of goal with tracking error < 10 meters) is clearly unattainable under extreme conditions. A relaxed success criterion (arrival within 5,000 meters of goal) better characterizes performance in the survival regime. Under this metric, the high wind scenario achieves partial success with final position error approximately 4,200 meters from the goal.

This relaxed success definition acknowledges that certain environmental conditions preclude precision navigation, and mission planners must accept degraded outcomes when operations extend beyond design limits. The 43% success

rate (within 5 km proximity) for high wind scenarios establishes quantitative boundaries for operational deployment, guiding go/no-go decisions based on weather forecasts and mission criticality.

E. Comparative Wind Analysis and Operational Envelope

1. Performance Metric Scaling

Table 1 summarizes quantitative performance across moderate and high wind conditions, revealing the dramatic non-linear degradation between operational regimes.

Table 2 Performance comparison across wind conditions

Metric	Moderate Wind (6-8 m/s)	High Wind (>20 m/s)
Average tracking error (m)	2.0	2,466
Degradation factor	1×	1,233×
Maximum deviation (m)	32	5,509
Degradation factor	1×	172×
Total integrated cost	9,584.7	2.10×10^{11}
Degradation factor	1×	2.19×10^7 ×
Path complexity (waypoints)	8	2
Reduction	0%	75%
Mission success (500m)	100%	0%
Mission success (5km)	100%	43%

The 1,233-fold increase in average tracking error and 21.9 million-fold increase in control cost demonstrate exponential rather than linear degradation. This non-linear scaling indicates threshold effects where crossing critical wind speed boundaries triggers regime changes in system behavior. The transition from controlled navigation to survival mode does not occur gradually; rather, performance collapses rapidly once environmental forcing exceeds control authority margins.

Path complexity reduction from 8 to 2 waypoints represents algorithmic adaptation to infeasible conditions, but this 75% simplification proves insufficient to restore nominal tracking performance. The mission success rate (within 500 meters) drops from 100% to 0%, while relaxed success (within 5 kilometers) degrades from 100% to only 43%. These statistics establish clear boundaries for reliable autonomous operation.

2. Bimodal Wind Distribution Effects

The operational environment exhibits bimodal wind distributions with distinct moderate and high wind modes. Bimodal analysis quantifies expected performance accounting for probabilistic occurrence of each regime. Under the moderate wind mode characterized by speeds 6-8 m/s with occurrence probability $P_{moderate} = 0.7$, the system achieves expected cost $J_{moderate} = 9,584.7$ and mission success rate 98%. The high wind mode with speeds exceeding 20 m/s and probability $P_{high} = 0.3$ results in expected cost $J_{high} = 2.1 \times 10^{11}$ and success rate 43% (within 5 km threshold).

Combined expected cost across both modes: $E[J] = 0.7 \times J_{moderate} + 0.3 \times J_{high} \approx 6.3 \times 10^{10}$, highlighting the disproportionate impact of high wind events on overall system performance. The 30% probability of encountering extreme conditions dominates the expected cost calculation, revealing that rare but severe events drive mission risk more than nominal operations.

Mission success probability combines mode-weighted outcomes: $P_{success} = 0.7 \times 0.98 + 0.3 \times 0.43 = 0.686 + 0.129 = 0.815$ (81.5% overall success rate within 5 km). This calculation reveals that despite excellent nominal performance, the system achieves only 81.5% reliability when accounting for worst-case environmental conditions. Risk-averse mission planning must accommodate this degraded success probability when operating in uncertain forecasted conditions.

3. Operational Boundaries Definition

Systematic performance characterization across wind speed regimes enables quantitative definition of operational boundaries for autonomous deployment:

Nominal operations ($w < 0.5 \times v_M^*$, or $w < 7$ m/s):

- 1) Tracking error: $e_{avg} < 10$ m, $e_{max} < 100$ m
- 2) Control cost: $J < 10^5$
- 3) Mission success: > 98
- 4) Status: Excellent performance, suitable for routine autonomous operations

Degraded operations ($0.5 \times v_M^* < w < v_M^*$, or 7-14 m/s):

- 1) Tracking error: $10 \text{ m} < e_{avg} < 1,000$ m
- 2) Control cost: $10^5 < J < 10^8$
- 3) Mission success: 80-95
- 4) Status: Acceptable with reduced accuracy, require increased monitoring

Emergency operations ($v_M^* < w < 1.5 \times v_M^*$, or 14-21 m/s):

- 1) Tracking error: $e_{avg} > 1,000$ m
- 2) Control cost: $J > 10^8$
- 3) Mission success: 40-80
- 4) Status: Survival mode, mission completion uncertain

Mission abort recommended ($w > 1.5 \times v_M^*$, or $w > 21$ m/s):

- 1) Control authority insufficient for reliable navigation
- 2) Mission success: < 40
- 3) Status: Operation outside design envelope, high risk of failure

These thresholds provide quantitative guidance for mission planning and go/no-go decisions based on weather forecasts. Conservative operations would limit autonomous missions to nominal conditions ($w < 7$ m/s), while risk-tolerant applications might accept degraded performance up to emergency thresholds (14 m/s) with appropriate contingency planning.

F. Information Gathering Under Degraded Conditions

1. SAA Performance Degradation

The SAA's information gathering capability degrades significantly under high wind conditions, limiting the collaborative benefits demonstrated in moderate scenarios. Spatial coverage reduces from 47 cells visited in moderate conditions to only 12 cells in high winds, representing a 74% reduction. Information gain rate decreases from 2.3 bits/second to 0.4 bits/second, an 83% reduction that severely limits the value of reconnaissance operations.

Effective sensing range contracts from 200 miles to approximately 50 miles as large tracking errors prevent targeted uncertainty sampling. Path deviations carry the SAA away from intended high-information regions, resulting in opportunistic rather than strategic observations. The degraded tracking performance fundamentally undermines the IT-RRT* algorithm's assumption that the vehicle can navigate to selected high-value cells.

2. Adaptive Sampling Strategies

Under high wind conditions, the SAA abandons optimal information path planning in favor of three survival-oriented strategies. First, opportunistic sampling along wind-feasible trajectories replaces strategic exploration; the SAA observes whatever cells it passes through rather than actively targeting high-information regions. Second, communication maintenance takes priority over information maximization to ensure the HVAA receives whatever data the SAA can gather before potential communication loss. Third, emergency return-to-base protocols activate when tracking error $\|e(t)\| > 3,000$ m, recognizing that continued reconnaissance under control-saturated conditions risks total SAA loss without commensurate information benefit.

3. HVAA Replanning Under Limited Information

With reduced SAA reconnaissance effectiveness, the HVAA receives information about only 23% of the planned corridor (versus 64% in moderate conditions). Replanning events decrease from 3 to 1 due to insufficient new information to justify dynamic route changes. The HVAA defaults to conservative RRT*-ETT worst-case planning, effectively

reverting to single-agent operation. Mission extension of additional 1,247 seconds results from conservative routing required when collaborative information gathering fails to provide adequate coverage.

The degraded collaborative performance highlights a critical system limitation: multi-agent coordination provides substantial benefits under nominal conditions but offers little advantage when environmental severity exceeds both agents' control capabilities. Mission planning must account for this correlation between information gathering effectiveness and environmental conditions, recognizing that reconnaissance is most valuable precisely when it is most difficult to execute.

G. Framework Robustness Assessment

1. Graceful Degradation Characteristics

The results demonstrate graceful degradation where the system maintains partial functionality under extreme stress rather than experiencing catastrophic failure. Path simplification from 8 to 2 waypoints, transition to goal-seeking mode, and mission completion despite severe tracking errors indicate adaptive responses that prioritize core objectives over secondary performance metrics.

This graceful degradation reflects three architectural features. First, the RRT* algorithm automatically simplifies paths when tree construction fails to find complex feasible solutions, providing built-in adaptation to constraint violations. Second, the control system implements emergency protocols that activate based on tracking error thresholds, enabling mode transitions without manual intervention. Third, the utility function formulation naturally trades information gain for path feasibility when environmental conditions tighten constraints.

2. System Limitations and Failure Modes

Three fundamental limitations prevent nominal performance restoration under extreme conditions. First, lack of intermediate recovery strategies between nominal and emergency modes results in abrupt performance cliffs rather than gradual degradation curves. The system offers only two operating points (excellent or survival) without intermediate degraded-but-functional states.

Second, information gathering becomes ineffective above critical wind thresholds, eliminating collaborative advantages precisely when additional information would be most valuable for safe navigation. The correlation between environmental severity and reconnaissance difficulty creates a negative feedback loop where uncertainty increases fastest when uncertainty reduction capability decreases most.

Third, exponential cost growth indicates need for improved high-wind control laws rather than mere parameter tuning. The current control architecture fundamentally lacks the authority to counteract extreme disturbances, requiring architectural modifications (adaptive control, robust control, or nonlinear disturbance rejection) rather than incremental improvements to existing algorithms.

3. Comparative Assessment

Despite limitations, the framework demonstrates clear superiority over alternative approaches within its design envelope. Compared to single-agent conservative planning, the collaborative approach achieves 28% path length reduction and 24.1% travel time improvement while maintaining equivalent safety margins. Compared to information-naïve exploration (random sampling), the entropy-based utility function provides 73% information utilization efficiency, indicating strategic rather than opportunistic reconnaissance.

The 2-meter average tracking accuracy under moderate conditions exceeds GPS-based navigation precision by an order of magnitude, validating the nonlinear control approach. The ability to maintain mission completion even under extreme degradation (43% success rate within 5 km) surpasses brittle systems that fail completely when environment exceeds nominal assumptions.

VI. Conclusion

This paper presents a novel information-theoretic multi-agent path planning framework for autonomous navigation under extreme environmental uncertainty. Through integration of entropy-based RRT* algorithms, vehicle dynamics constraints, and nonlinear trajectory tracking control, the framework enables collaborative sensing strategies that substantially improve mission performance while maintaining safety requirements.

A. Key Contributions and Results

The collaborative framework demonstrates significant performance improvements over conservative single-agent approaches. Simulation results using Hurricane Harvey 2017 wind data validate 28% path length reduction and 24.1% travel time improvement when SAA reconnaissance enables HVAA route optimization. Information utilization efficiency reaches 73%, with collaborative gain factor of 1.34 quantifying the multiplicative benefit of multi-agent coordination.

Trajectory tracking under moderate wind conditions (6-8 m/s) achieves exceptional precision with 2.0-meter average cross-track error and 32-meter maximum deviation during aggressive maneuvers. The information-theoretic utility function enables strategic exploration accumulating 847.3 bits at 2.3 bits per second, with entropy-based weighting ensuring reconnaissance targets high-value uncertain regions. These results establish baseline capability exceeding GPS-based navigation precision by an order of magnitude.

B. Operational Boundaries and Limitations

Comprehensive performance characterization reveals well-defined operational boundaries. Nominal operations below 7 m/s wind speed maintain tracking errors under 10 meters with 98% mission success. Degraded operations at 7-14 m/s exhibit acceptable performance for many applications. Emergency operations at 14-21 m/s result in tracking errors exceeding 1,000 meters with 40-80% success rates. Operations above 21 m/s fall outside the design envelope.

High wind conditions expose fundamental limitations. The 1,233-fold increase in tracking error and 21.9 million-fold increase in control cost demonstrate exponential degradation revealing threshold effects. Information gathering effectiveness decreases 74% in spatial coverage and 83% in acquisition rate, eliminating collaborative advantages when environmental severity exceeds control capabilities. The lack of intermediate recovery strategies results in abrupt performance cliffs between nominal and survival modes.

Despite these limitations, the framework maintains graceful degradation with 43% mission completion under extreme stress, demonstrating robustness essential for autonomous systems in uncertain environments.

C. Implications and Future Work

The quantified boundaries provide actionable guidance for operational deployment. Risk-averse applications should limit operations to nominal conditions (wind < 7 m/s), while risk-tolerant missions might accept degraded performance to 14 m/s with appropriate contingencies. The 28% efficiency improvement justifies multi-agent deployment for time-critical missions including disaster response and environmental monitoring.

Future research directions include Thread 2 implementation of indirect information gain through spatial correlation, potentially reducing reconnaissance coverage by 40-60% while maintaining information value. Thread 3 addresses heterogeneous risk source parameterization, specifically wind-threat coupling effects. Advanced control strategies for high-wind operation—including adaptive control, robust control, and model predictive approaches—represent critical needs for extending operational boundaries.

Experimental validation through hardware-in-the-loop simulation and flight testing will assess performance pattern transfer from simulation to physical systems, calibrating confidence for operational deployment.

The information-theoretic multi-agent framework successfully demonstrates that collaborative information gathering enables substantially more efficient navigation through uncertain environments. The framework achieves design objectives within nominal envelope while gracefully degrading under stress. Honest characterization of both capabilities and limitations, quantified operational boundaries, and identified technical gaps provide measured progress toward autonomous systems reliably operating in severe conditions. By establishing realistic expectations about current state-of-the-art performance, this work contributes to incremental advancement of autonomous navigation technology for hazardous environment operations.

VII. Appendix

Algorithm 1 HVAA RRT*-ETT

```

1: function HVAA RRT*-ETT
2:    $T \leftarrow \text{InitializeTree}()$ 
3:    $T \leftarrow \text{InitializeNode}(\emptyset, q_{\text{init}}, T)$ 
4:   for  $i = 1$  to  $N$  do
5:      $q_{\text{rand}} \leftarrow \text{RandomSample}(i)$ 
6:      $q_{\text{nearest}} \leftarrow \text{NearestETT}(T, q_{\text{rand}})$ 
7:      $q_{\text{new}}, U_{\text{new}} \leftarrow \text{Steer}(q_{\text{nearest}}, q_{\text{rand}})$ 
8:     if  $\text{NoExceed}(q_{\text{nearest}}, q_{\text{rand}})$  then
9:        $q_{\text{near}} \leftarrow \text{Near}(T, q_{\text{new}})$ 
10:       $q_{\text{min}}, U_{\text{min}} \leftarrow \text{ParentETT}(q_{\text{near}}, q_{\text{nearest}}, \dots, q_{\text{new}}, U_{\text{new}})$ 
11:    end if
12:  end for
13: end function

```

Algorithm 2 NearestETT

```

1: function NEARESTETT( $T, q_{\text{rand}}$ )
2:   for  $i = 1$  to  $N$  do
3:      $\text{TempTT}(i) = \text{TravelTime}(T(i), q_{\text{rand}})$ 
4:   end for
5:   return  $\text{Min}(\text{TempTT})$ 
6: end function

```

Algorithm 3 ParentETT

```

1: function PARENTETT( $q_{\text{near}}, q_{\text{nearest}}, \dots, q_{\text{new}}, U_{\text{new}}$ )
2:    $\triangleright U_{\text{min}} = U_{\text{new}}$ 
3:   for  $i = 1$  to  $N$  do
4:     if  $\text{TravelTime}(q_{\text{new}}, q_{\text{near}}(i)) < U_{\text{min}}$  then
5:        $q_{\text{min}} \leftarrow q_{\text{near}}(i)$ 
6:        $U_{\text{min}} \leftarrow \text{TravelTime}(q_{\text{new}}, q_{\text{near}})$ 
7:     else
8:        $\triangleright \text{Continue}$ 
9:     end if
10:  end for
11:  return  $q_{\text{min}}, U_{\text{min}}$ 
12: end function

```

Algorithm 4 NoExceed

```
1: function NoEXCEED( $q_{\text{new}}, q_{\text{near}}$ )
2:    $\text{path} \leftarrow \text{DiscretizePath}(q_{\text{new}}, q_{\text{near}})$ 
3:   for  $i < \text{Length}(\text{path})$  do
4:     if  $\text{path}(i) < \text{MaxSD} \vee \text{MaxRisk}(\lambda^{\text{HVAA}})$  then
5:        $\triangleright$  Continue
6:     else
7:       return Exceed = 1
8:        $\triangleright$  Break
9:     end if
10:  end for
11:  return Exceed = 0
12: end function
```

Algorithm 5 SAA RRT*-IT

```
1: function SAA RRT*-IT
2:    $T \leftarrow \text{InitializeTree}()$ 
3:    $T \leftarrow \text{InitializeNode}(\emptyset, q_{\text{init}}, T)$ 
4:   for  $i = 1$  to  $N$  do
5:      $q_{\text{rand}} \leftarrow \text{RandomSample}(i)$ 
6:      $q_{\text{nearest}} \leftarrow \text{NearestIT}(T, q_{\text{rand}})$ 
7:      $q_{\text{new}}, U_{\text{new}} \leftarrow \text{Steer}(q_{\text{nearest}}, q_{\text{rand}})$ 
8:     if NoExceed( $q_{\text{nearest}}, q_{\text{rand}}$ ) then
9:        $q_{\text{near}} \leftarrow \text{Near}(T, q_{\text{new}})$ 
10:       $q_{\text{min}}, U_{\text{min}} \leftarrow \text{ParentIT}(q_{\text{near}}, q_{\text{nearest}}, q_{\text{new}}, U_{\text{new}})$ 
11:       $T \leftarrow \text{InsertNode}(q_{\text{min}}, q_{\text{new}}, T)$ 
12:       $T \leftarrow \text{Rewire}(T, q_{\text{near}}, q_{\text{min}}, q_{\text{new}})$ 
13:    end if
14:  end for
15: end function
```

Algorithm 6 NearestIT

```
1: function NEARESTIT( $T, q_{\text{rand}}$ )
2:   for  $i = 1$  to  $N$  do
3:      $\text{TempU}(i) = \text{Utility}(T(i), q_{\text{rand}})$ 
4:   end for
5:   return Min(TempU)
6: end function
```

Algorithm 7 ParentIT

```
1: function PARENTIT( $q_{\text{near}}, q_{\text{nearest}}, q_{\text{new}}, U_{\text{new}}$ )
2:    $\triangleright U_{\text{min}} = U_{\text{new}}$ 
3:   for  $i = 1$  to  $N$  do
4:     if  $\text{Utility}(q_{\text{new}}, q_{\text{near}}(i)) < U_{\text{min}}$  then
5:        $q_{\text{min}} \leftarrow q_{\text{near}}(i)$ 
6:        $U_{\text{min}} \leftarrow \text{Utility}(q_{\text{new}}, q_{\text{near}})$ 
7:     else
8:        $\triangleright$  Continue
9:     end if
10:  end for
11:  return  $q_{\text{min}}, U_{\text{min}}$ 
12: end function
```

Algorithm 8 NoExceed (SAA)

```
1: function NoEXCEED( $q_{\text{new}}, q_{\text{near}}$ )
2:   path  $\leftarrow$  DiscretizePath( $q_{\text{new}}, q_{\text{near}}$ )
3:   for  $i < \text{Length}(\text{path})$  do
4:     if path( $i$ )  $< \text{MaxSD} \wedge \text{MaxRisk}(\lambda^{\text{SAA}})$  then
5:        $\triangleright$  Continue
6:     else
7:       return Exceed = 1
8:        $\triangleright$  Break
9:     end if
10:  end for
11:  return Exceed = 0
12: end function
```

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